Geological Cobalt prospectivity in the Ruvuma region, to include details of geology and its favorable environment.

Mineral: Cobalt

Region: Ruvuma

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Regional Geological Setting

Geological Features:

- **Mafic-ultramafic Rocks:** Deep mafic layers often include large deposits of copper and nickel.
- **Stratiform Copper Deposits:** Copper deposits in stratified regions such as the Central Lapland Belt, Peräpohja Belt, Kuusamo Belt, Hitura region, and Outokumpu region.

Environmental Conditions:

- Low exposure to water and oxygen
- Access to favorable soil conditions
- High mineral availability (e.g., cobalt)

Mineral Potential

Cobalt is a potentially critical mineral that has substantial potential for long-term production in the Ruvuma region. The exploration history and infrastructure areas provide an extensive base for exploration, including large-scale mining operations, research facilities, and exploration permits.

Exploration History:

- **Central Lapland Belt:** Initial exploration and development was focused on 1,000 metric tons of cobalt per year of contained.
- **Peräpohja Belt:** Expansion efforts continued with exploration in the region, which resulted in a combined model covering low prospectivity areas and moderate to high prospectivity areas.
- **Kuusamo Belt:** Additional mining activities developed this area, particularly focusing on the central Lapland belt.
- **Hitura region:** Exploration was continued in the Hitura belt and Outokumpu region.
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Infrastructure Areas:

- **Central Lapland Belt:** Includes significant exploration areas such as the Central Lapland Belt, Peräpohja Belt, Kuusamo Belt, Hitura region, and Outokumpu region.
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Risks:
1. **High Potential for Low Proximity:** The exploration area in Ruvuma region is highly vulnerable to low proximity, which can lead to exploration permits being granted to inactive explorers.
2. **Resource Limitations:** Limited exploration areas result in high mineral availability within the region, potentially limiting the production of cobalt.
3. **Technological Challenges:** Significant technical challenges may hinder exploration and development.

Recommendations:
1. **Investigate Exploration Areas:** Conduct extensive exploration in Ruvuma to identify potential areas for active exploration.
2. **Optimize Exploration Areas:** Design and optimize exploration areas to reduce exposure limits, such as reducing water depth, increasing oxygen availability, or using more efficient technologies.
3. **Support Research and Development:** Encourage research and development of new mining techniques, including the development of advanced techniques like the tectonic environments and the use of evaporites in the section.
4. **Enhance Exploration Permit Areas:** Increase the number of exploration permits available in Ruvuma region to ensure that active explorers have access to favorable environments.

Geographic Coverage

The Ruvuma region covers both central Lapland Belt, Peräpohja Belt, Kuusamo Belt, Hitura region, and Outokumpu region. This is a well-documented exploration area with significant potential for active exploration.

Map Visualization

Central Lapland Belt: Open circles highlight the areas that have active exploration for cobalt based on exploration permits.
Peräpohja Belt: Similar to the Central Lapland Belt, these areas are also indicated as favourable for cobalt exploration in the prospectivity modelling conducted by Nykänen _et al_.
Kuusamo Belt: This area has potential for Kuusamo-type deposits. According to the Outokumpu-type model, this deposit type is favourable not only within the Outokumpu region, but also within central Lapland.
Hitura region: This area includes significant exploration areas and development of new mining facilities.

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- Limited access to water resources within the Ruvuma region.
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Access to Favorable Soil Conditions

- Lower exposure to soil conditions within the Ruvuma region.
- Moderate to locally massive sulfides within deformed siliciclastic metasedimentary rocks.

High Mineral Availability

- Significant mineral availability in the Ruvuma region, which is highly critical for long-term production of cobalt.

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References and Sources

- Cobalt Prospectivity Using a Conceptual Fuzzy Logic Overlay Method Enhanced with the Mineral Systems Approach | Natural Resources Research | Springer Nature Link
- [PDF] Cobalt Mineral Exploration and Supply From 1995 Through 2013
- Cobalt, Styles of deposits and the search for primary deposits - USGS
- A mining industry overview of cobalt in Finland: exploration, deposits and utilization | Geoenergy
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Key findings

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